

ADVANCED MATHEMATICAL PHYSICS

PHYS 721

11.3.3 Show that the integral

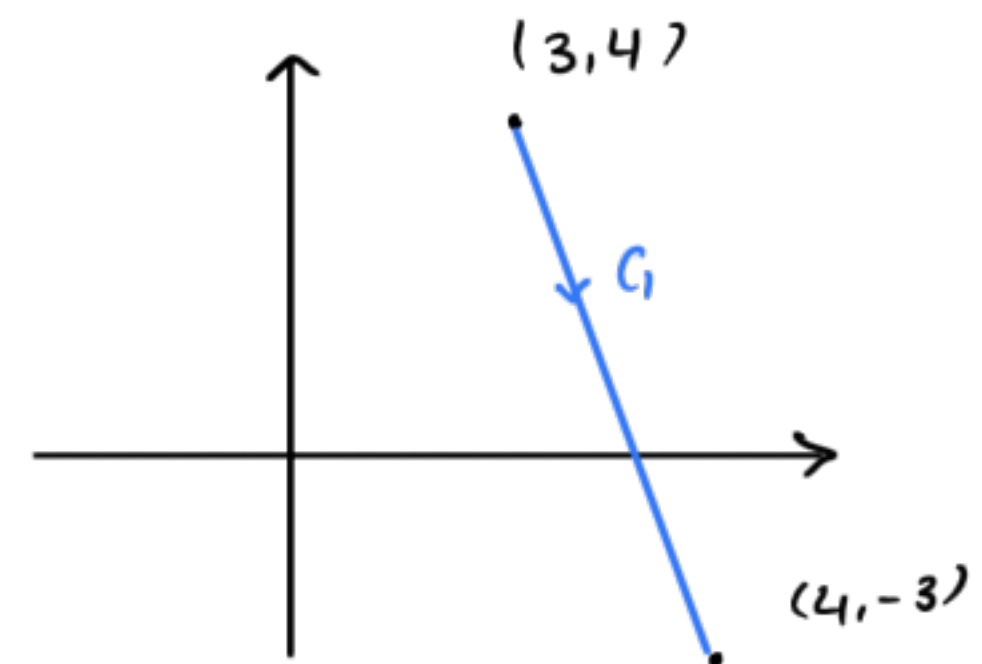
$$\int_{3+4i}^{4-3i} (4z^2 - 3iz) dz$$

has the same value on the two paths: (a) the straight line connecting the integration limits, and (b) an arc on the circle $|z| = 5$.

Goal : Evaluate $\int_{3+4i}^{4-3i} 4z^2 - 3iz dz$ on two paths

Path 1 :

$$\begin{aligned} dz &= dz \\ \int_{3+4i}^{4-3i} 4z^2 - 3iz dz &= \left. \frac{4z^3}{3} - \frac{3iz^2}{2} \right|_{3+4i}^{4-3i} \\ &= \frac{4}{3}(4-3i)^3 - \frac{3i}{2}(4-3i)^2 - \left[\frac{4}{3}(3+4i)^3 + \frac{3i}{2}(3+4i)^2 \right] \quad (1) \\ &= \frac{76}{3} - \frac{707}{3}i \end{aligned}$$

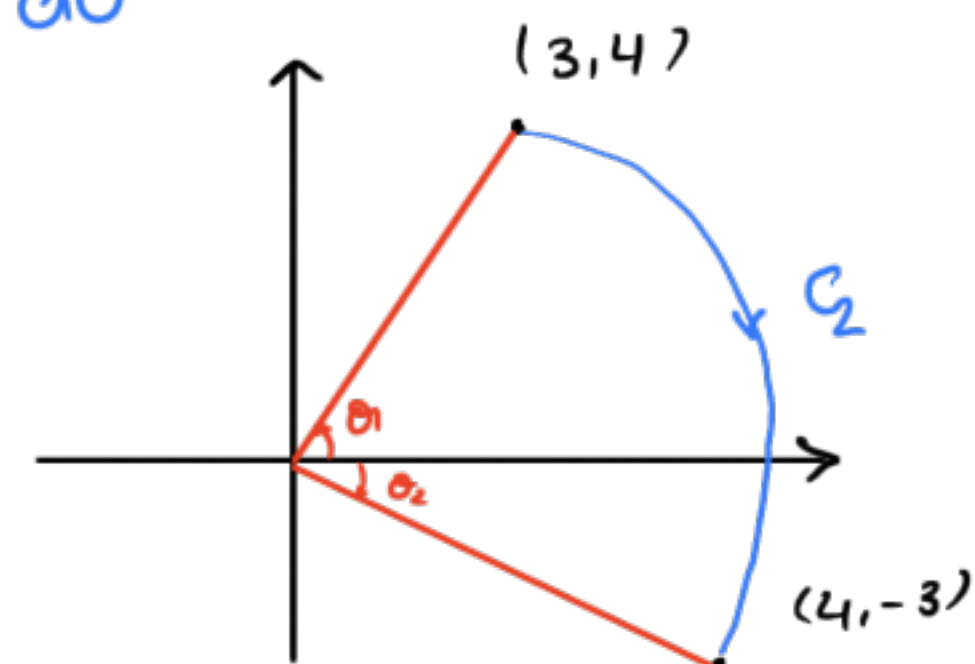


Path 2 :

We parametrize C_2 using the polar coordinates.

We have that r is constant with $r=5$. $dz = ire^{i\theta} d\theta = 5ie^{i\theta} d\theta$

$$\begin{aligned} \int_{3+4i}^{4-3i} 4z^2 - 3iz dz &= \int_{\theta_2}^{\theta_1} [100e^{i2\theta} - 15ie^{i\theta}] 5ie^{i\theta} d\theta \\ &= \int_{\theta_2}^{\theta_1} 500ie^{i3\theta} + 75e^{i2\theta} d\theta \\ &= \left. -\frac{500}{3}e^{i3\theta} + \frac{75i}{2}e^{i2\theta} \right|_{\theta_2}^{\theta_1} \\ &= -\frac{500}{3} \frac{(3+4i)^3 - (4-3i)^3}{5^3} + \frac{75i}{2} \frac{(3+4i)^2 - (4-3i)^2}{5^2} \\ &= \frac{76}{3} + \frac{707}{3}i \end{aligned}$$



→ This is exactly as (1) !

$$\begin{aligned} (e^{i\theta_1})^3 &= \left(\frac{3+4i}{5}\right)^3, & (e^{i\theta_1})^2 &= \left(\frac{3+4i}{5}\right)^2 \\ (e^{i\theta_2})^3 &= \left(\frac{4-3i}{5}\right)^3, & (e^{i\theta_2})^2 &= \left(\frac{4-3i}{5}\right)^2 \end{aligned}$$

11.3.6 Verify that

$$\int_0^{1+i} z^* dz$$

depends on the path by evaluating the integral for the two paths shown in Fig. 11.7. Recall that $f(z) = z^*$ is not an analytic function of z and that Cauchy's integral theorem therefore does not apply.

Goal is Evaluate $\int_0^{1+i} z^* dz$ on different paths

Path 1 is

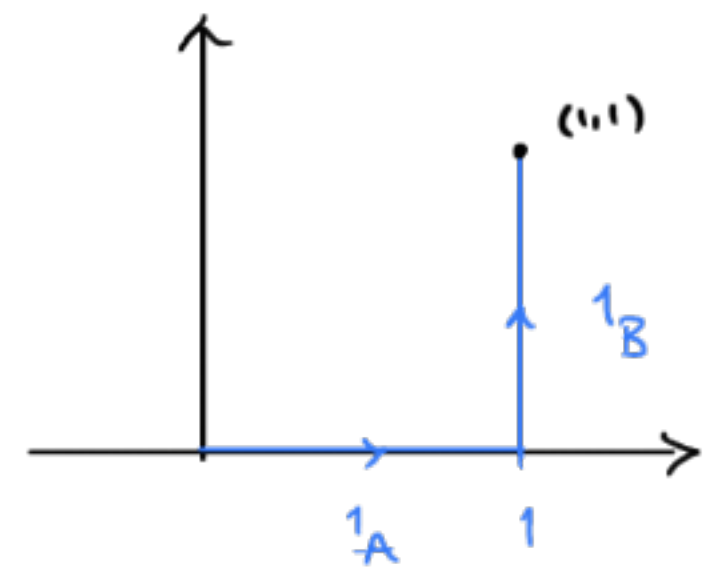
Segment 1A is $dz = dx, y=0$

$$\int_0^1 x dx = \frac{x^2}{2} \Big|_0^1 = \frac{1}{2}$$

Segment 1B is $dz = dy, x=1$

$$\int_0^1 [1-iy] i dy = i y + \frac{y^2}{2} \Big|_0^1 = i + \frac{1}{2}$$

$$\int_0^{1+i} z^* dz = \frac{1}{2} + i + \frac{1}{2} = 1+i$$



Path 2 is

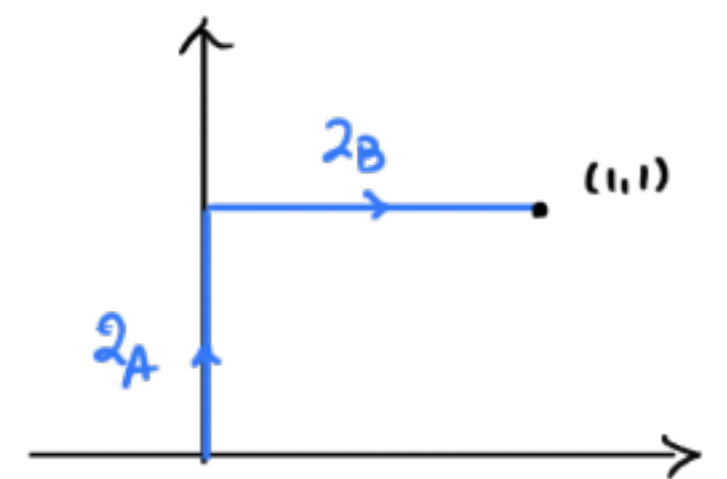
Segment 2A is $dz = dy, x=0$

$$\int_0^1 [-iy] i dy = \frac{y^2}{2} \Big|_0^1 = \frac{1}{2}$$

Segment 2B is $dz = dx, y=1$

$$\int_0^1 x - i dx = \frac{x^2}{2} - ix \Big|_0^1 = \frac{1}{2} - i$$

$$\int_0^{1+i} z^* dz = \frac{1}{2} + \frac{1}{2} - i = 1-i$$



* We note that different contours yield different integral answers since z^* is **NOT** an analytic function of z .

11.3.7 Show that

$$\oint_C \frac{dz}{z^2 + z} = 0,$$

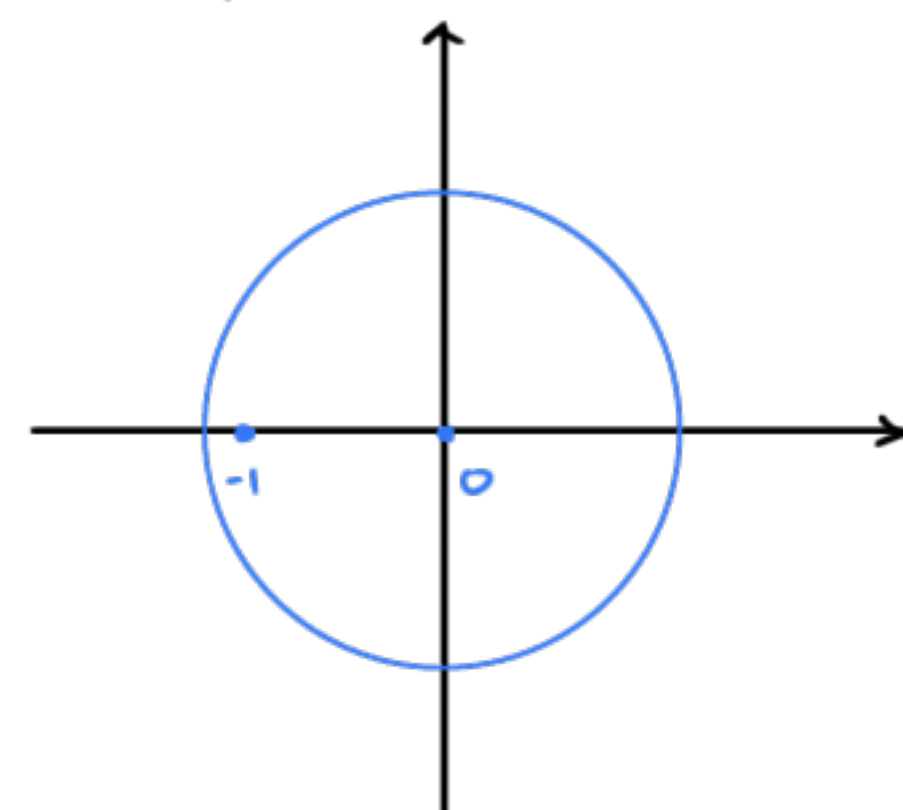
in which the contour C is a circle defined by $|z| = R > 1$.

Hint. Direct use of the Cauchy integral theorem is illegal. The integral may be evaluated by expanding into partial fractions and then treating the two terms individually. This yields 0 for $R > 1$ and $2\pi i$ for $R < 1$.

Goal Show that $\oint_C \frac{dz}{z^2 + z} = 0$ for $|z| = R > 1$

Proof

$$\oint_C \frac{dz}{z^2 + z} = \oint_C \frac{1}{z} - \frac{1}{z+1} dz = 0$$



By this,

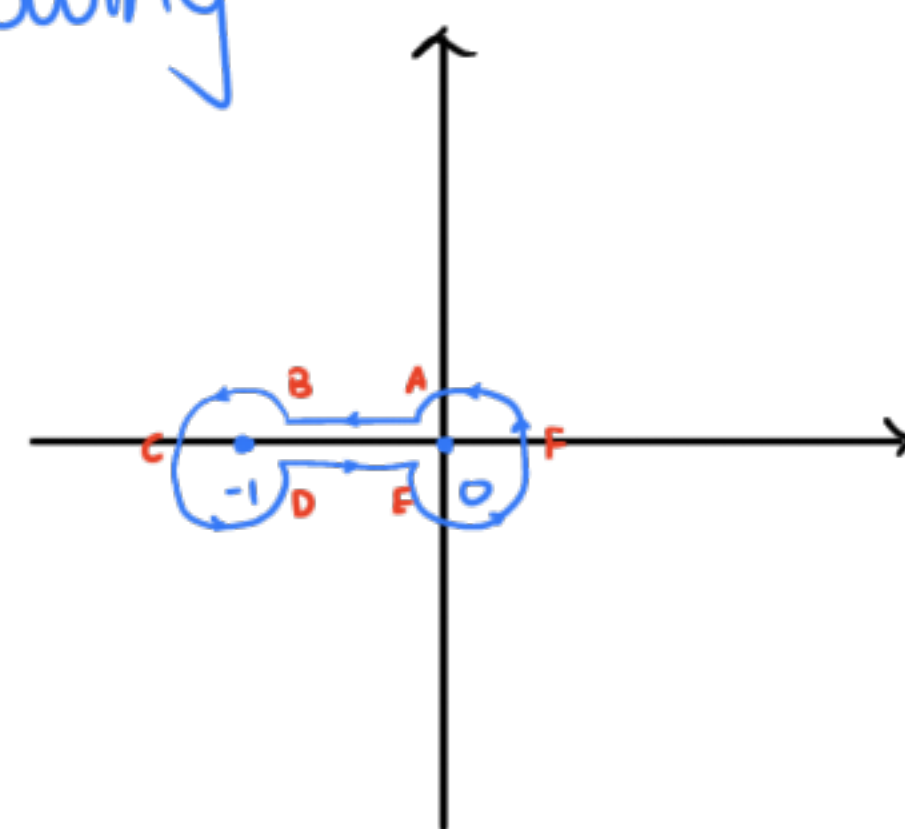
The integral of an analytic function over a closed path has a value that remains unchanged over all possible continuous deformations of the contour within the region of analyticity.

We can transform the contour C as the following

We have now split our new contour into 4 segments

$$\text{s.t. } \oint_C \frac{dz}{z^2 + z} = \int_{AB} \dots dz + \int_{BCD} \dots dz + \int_{DE} \dots dz + \int_{EFA} dz$$

are equal but opposite in direction & so cancel.



$$= \int_{BCD} \frac{1}{z} - \frac{1}{z+1} dz + \int_{EFA} \frac{1}{z} - \frac{1}{z+1} dz$$

singularity $z=0$ is outside the region enclosed by BCD
 $\therefore \int \frac{1}{z} dz = 0$

singularity $z=-1$ is outside the region enclosed by EFA
 $\therefore \int \frac{-1}{z+1} dz = 0$

$$= \int_{BCD} - \frac{1}{z+1} dz + \int_{EFA} \frac{1}{z} dz$$

$$= -2\pi i + 2\pi i \quad \text{by 11.29}$$

$$= 0$$

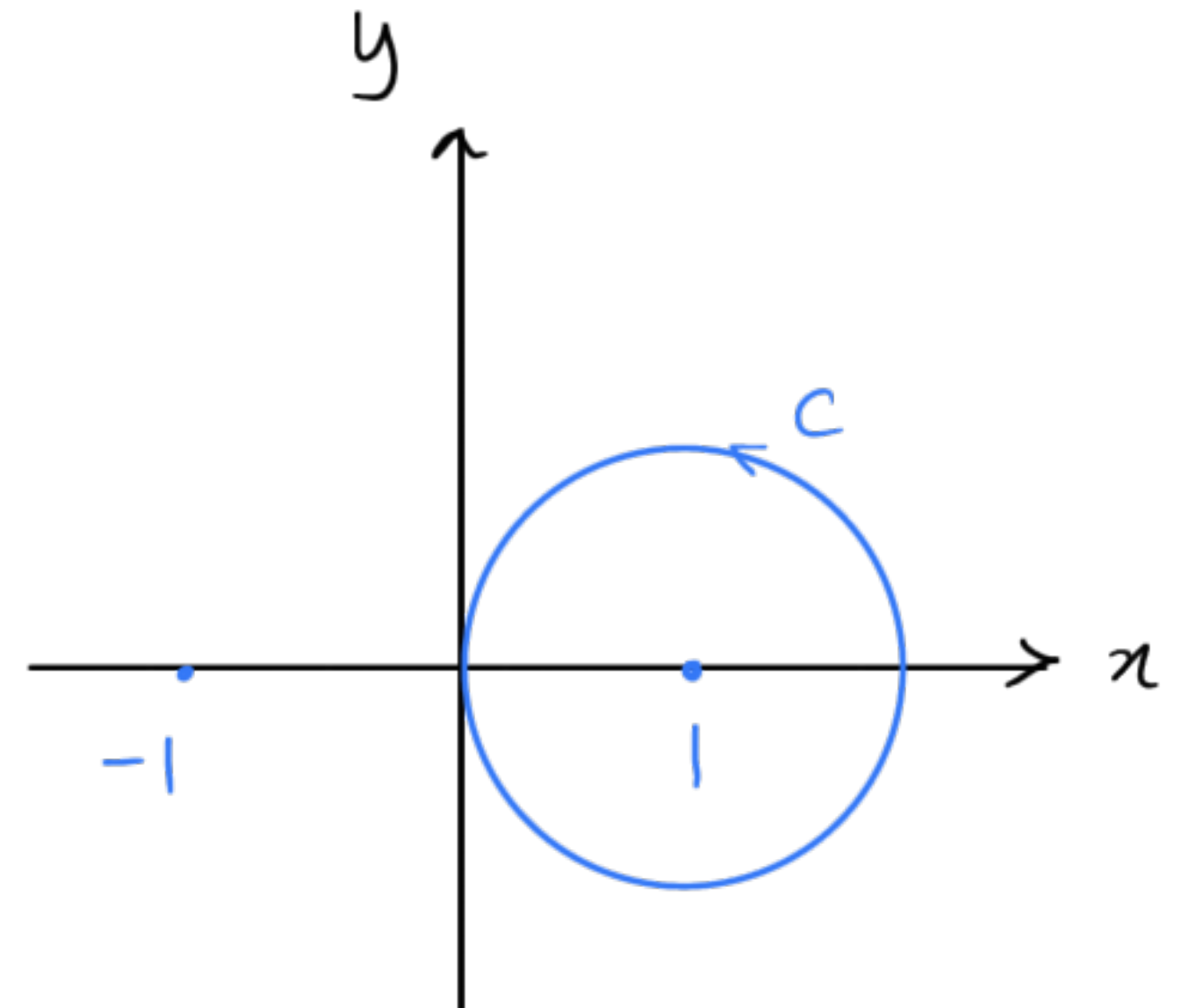
11.4.2 Evaluate

$$\oint_C \frac{dz}{z^2 - 1},$$

where C is the circle $|z - 1| = 1$.

Solution

$$\oint_C \frac{dz}{z^2 - 1} = \frac{1}{2} \oint_C \frac{1}{z - 1} - \frac{1}{z + 1} dz$$



singularity $z = -1$
out side region enclosed
by $C \Rightarrow \oint_C \frac{1}{z+1} dz = 0$

$$= \frac{1}{2} 2\pi i = \pi i$$

by Cauchy's Integral
Formula.

11.4.6 Evaluate

$$\oint_C \frac{e^{iz}}{z^3} dz,$$

for the contour a square with sides of length $a > 1$, centered at $z = 0$.

Solution

Contour is irrelevant. We can use 11.33 to find this since the singularity $z=0$ is enclosed in C .

$$f^{(n)}(z_0) = \frac{n!}{2\pi i} \int \frac{f(z)}{(z - z_0)^{n+1}} dz$$

$$\oint_C \frac{e^{iz}}{z^3} dz = \frac{2\pi i}{2!} \left[\frac{d^2}{dz^2} e^{iz} \right]_{z=0}$$

$$= \pi i (-e^{iz}) \Big|_{z=0} = -\pi i$$

11.4.7 Evaluate

$$\oint_C \frac{\sin^2 z - z^2}{(z-a)^3} dz,$$

where the contour encircles the point $z = a$.

Solution

The singularity $z=a$ is in the region enclosed by contour C .

Again, we can use 11.33

$$f^{(n)}(z_0) = \frac{n!}{2\pi i} \int \frac{f(z)}{(z-z_0)^{n+1}} dz \quad \text{to evaluate.}$$

$$\begin{aligned} \oint_C \frac{\sin^2 z - z^2}{(z-a)^3} &= \frac{2\pi i}{2!} \left. \frac{d}{dz^2} (\sin^2 z - z^2) \right|_{z=a} \\ &= \pi i \left. \frac{d}{dz} (2\sin z \cos z - 2z) \right|_{z=a} \\ &= \pi i \left[-2\sin^2 z + 2\cos^2 z - 2 \right] \Big|_{z=a} \\ &= 2\pi i (\cos 2z - 1) \Big|_{z=a} \\ &= 2\pi i (\cos 2a - 1) \end{aligned}$$

11.4.8 Evaluate

$$\oint_C \frac{dz}{z(2z+1)},$$

for the contour the unit circle.

Solution

Both singularities $z_1 = 0$ & $z_2 = -\frac{1}{2}$ is in the region enclosed by C . We can split both singularities by partial fraction & evaluate.

$$\begin{aligned} \oint_C \frac{1}{z(2z+1)} dz &= \oint_C \frac{1}{z} - \frac{2}{2z+1} dz = \oint_C \frac{1}{z} - \frac{1}{z-\frac{1}{2}} dz \\ &= 2\pi i - 2\pi i = 0 \end{aligned}$$

$$1 = A(2z+1) + Bz$$

$\boxed{A=1}$ $B=-2$

↑ Cauchy
Integral
formula.

11.4.9 Evaluate

$$\oint_C \frac{f(z)}{z(2z+1)^2} dz,$$

for the contour the unit circle.

Hint. Make a partial fraction expansion.

Solution

Both singularities $z_1 = 0$ & $z_2 = -\frac{1}{2}$ is in the region enclosed by C . We can split both singularities by partial fraction & evaluate

$$f^{(n)}(z_0) = \frac{n!}{2\pi i} \int \frac{f(z)}{(z-z_0)^{n+1}} dz$$

$$\oint_C \frac{f(z)}{z(2z+1)^2} = \oint_C \frac{f(z)}{z} dz - 2 \oint_C \frac{f(z)}{2z+1} - 2 \oint_C \frac{f(z)}{(2z+1)^2}$$

$$= \oint_C \frac{f(z)}{z} dz - \oint_C \frac{f(z)}{z + \frac{1}{2}} - \frac{1}{2} \oint_C \frac{f(z)}{(z + \frac{1}{2})^2}$$

$$= 2\pi i f(0) - 2\pi i f(-\frac{1}{2}) - \pi i f'(-\frac{1}{2})$$

11.5.1 Develop the Taylor expansion of $\ln(1+z)$.

$$\text{ANS. } \sum_{n=1}^{\infty} (-1)^{n-1} \frac{z^n}{n}.$$

Goal & Find Taylor Series expansion of $\ln(1+z)$

proof

We can use the antiderivative of the analytic continuation of the geometric series to find $\ln(1+z)$.

$$\frac{1}{1+z} = \frac{1}{1-(-z)} = \sum_{n=0}^{\infty} (-z)^n$$

↙ $\int dz$

$$\ln(1+z) = \int \left[\sum_{n=0}^{\infty} (-1)^n z^n \right] dz = \int [1 - z + z^2 - \dots] dz$$

$$= z - \frac{z^2}{2} + \frac{z^3}{3} - \dots$$

$$= \sum_{n=1}^{\infty} (-1)^{n-1} \frac{z^n}{n!}$$

11.5.6 Obtain the Laurent expansion of e^z/z^2 about $z = 0$.

Solution

$$\frac{e^z}{z^2} = \frac{\sum_{n=0}^{\infty} \frac{z^n}{n!}}{z^2} = \sum_{n=0}^{\infty} \frac{z^{n-2}}{n!} = \sum_{n=-2}^{\infty} \frac{z^n}{(n+2)!}$$

11.5.7 Obtain the Laurent expansion of $ze^z/(z-1)$ about $z = 1$.

Solution

$$\frac{ze^z}{z-1} = \frac{(z-1+1)e^{z-1} \cdot e}{z-1}$$

$$= e \cdot e^{z-1} \left[1 + \frac{1}{z-1} \right]$$
$$= e \sum_{n=0}^{\infty} \frac{(z-1)^n}{n!} \left(1 + \frac{1}{z-1} \right)$$

$$= e \left[\sum_{n=0}^{\infty} \frac{(z-1)^n}{n!} + \sum_{n=0}^{\infty} \frac{(z-1)^{n-1}}{n!} \right]$$

$$= e \left[\sum_{n=0}^{\infty} \frac{(z-1)^n}{n!} + \sum_{n=-1}^{\infty} \frac{(z-1)^n}{(n+1)!} \right]$$

$$= e \left[\frac{1}{z-1} + \sum_{n=0}^{\infty} \frac{(z-1)^n}{n!} + \sum_{n=0}^{\infty} \frac{(z-1)^n}{(n+1)!} \right]$$

$$= e \left[\frac{1}{z-1} + \sum_{n=0}^{\infty} (z-1)^n \frac{n+2}{(n+1)!} \right]$$

11.6.5 Find all singularities of

$$z^{-1/3} + \frac{z^{-1/4}}{(z-3)^3} + (z-2)^{1/2},$$

and identify their types (e.g., second-order branch point, fifth-order pole, ...). Include any singularities at the point at infinity.

Note. A branch point is of n th order if it requires n , but no fewer, circuits around the point to restore the original value.

Goal & Find all singularities of

$$z^{-1/3} + \frac{z^{-1/4}}{(z-3)^3} + (z-2)^{1/2}$$

Solution &

$$z^{-1/3} + \frac{z^{-1/4}}{(z-3)^3} + (z-2)^{1/2}$$

s_1 (pointing to $z^{-1/3}$) s_2 (pointing to $z^{-1/4}$) s_3 (pointing to $(z-3)^3$) s_4 (pointing to $(z-2)^{1/2}$)

s_3 & Singularity at $z=3$ of the third order. $\lim_{z \rightarrow 3} \frac{(z-3)^{\textcircled{3}}}{(z-3)^3} = 1$

s_4 & Square root f^n indicates a branch point of the second order at $z=2$

s_1 and s_2 & Both indicate a singularity at $z=0$.
When does the f^n return to its original value?

$$z^{-1/3} \curvearrowright 3 \text{ turns} \quad z^{-1/4} \curvearrowright 4 \text{ turns}$$

$\curvearrowright 3 \times 4 = 12 \text{ turns} \Rightarrow \text{branch point of order 12.}$

For $z \rightarrow \infty$, we examine ...

$$w^{1/3} \curvearrowright 3 \text{ turns} \quad w^{-1/4} \curvearrowright 4 \text{ turns} \quad w^{-1/2} \curvearrowright 2 \text{ turns}$$

$\curvearrowright 12 \text{ turns}$

$$w^{1/3} + \frac{w^{1/4}}{(\frac{1}{w}-3)^3} + (\frac{1}{w}-2)^{1/2}$$

$$w^{1/3} + \frac{w^{-1/4}}{(1-3w)^3} + w^{-1/2}(1-2w)^{1/2}$$

$z \rightarrow \infty \Rightarrow 12^{\text{th}} \text{ BP}$

11.6.6 The function $F(z) = \ln(z^2 + 1)$ is made single-valued by straight-line branch cuts from $(x, y) = (0, -1)$ to $(-\infty, -1)$ and from $(0, +1)$ to $(0, +\infty)$. See Fig. 11.17. If $F(0) = -2\pi i$, find the value of $F(i - 2)$.

Goal : find $F(i-2)$ for $F(z) = \ln(z^2 + 1)$
through branch cuts in fig. 11.17 & $F(0) = -2\pi i$

proof

Since the natural log function has infinite branches, we need to identify for which branch $F(0) = -2\pi i$ lies.

$$F(0) = \ln(0^2 + 1) = \ln 1 = \ln 1 + i(\pi + 2\pi n) = -2\pi i$$

$\Rightarrow \boxed{n = -1}$
The $n = -1$ branch!

$$\begin{aligned} \therefore F(i-2) &= \ln((i-2)^2 + 1) \\ &= \ln(-1 - 4i + 4 + 1) \\ &= \ln(4 - 4i) = \ln|4 - 4i| + i(-\pi/4 + 2\pi(-1)) \\ &= \ln 4\sqrt{2} - 9\pi/4 i \quad \square \end{aligned}$$

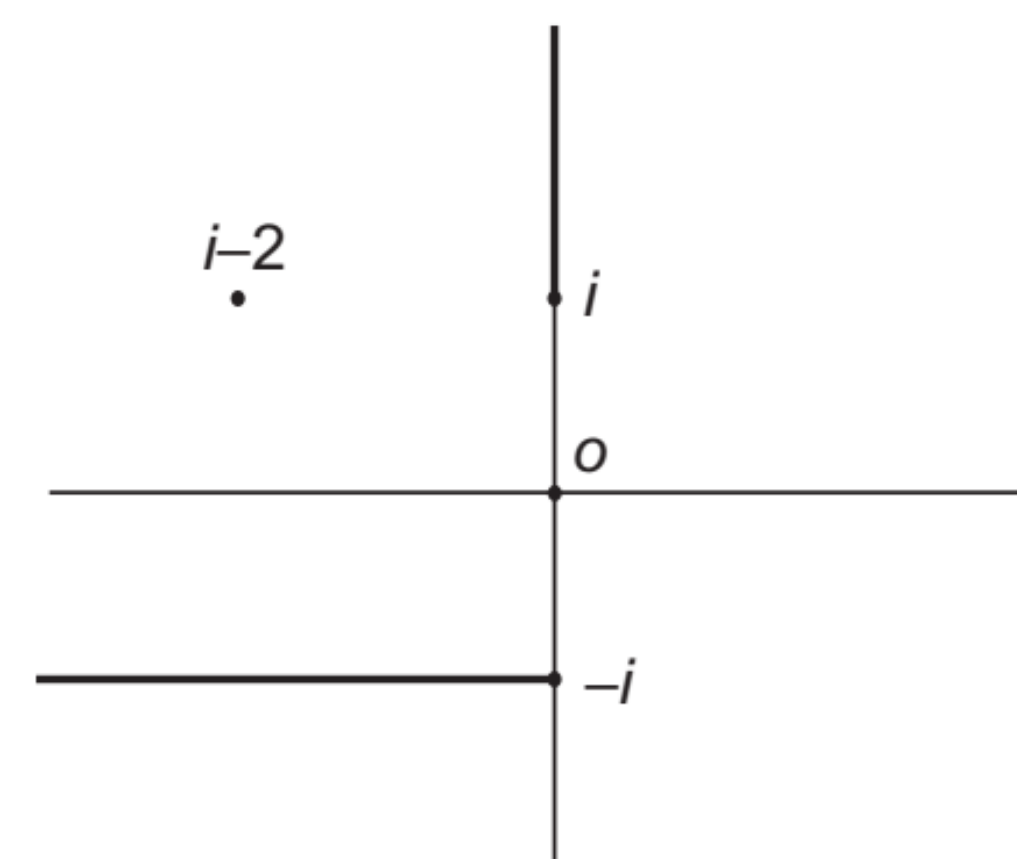


FIGURE 11.17 Branch cuts for Exercise 11.6.6.